

yolov10n Final Training Report

System Information

OS	Linux 6.6.105+
Python Version	3.12.12
PyTorch Version	2.9.0+cu126
CUDA Available	True
Device	cuda
RAM (GB)	13.61

Dataset Information

Property	Value
Dataset	bdd100k_yolo_limited
Number of Classes	10
Classes	pedestrian, rider, car, truck, bus, train, motorcycle, bicycle, traffic light, traffic sign
Train Images	945
Val Images	611
Test Images	725
Data YAML	data.yaml

Optimization Summary

Metric	Value
Total Trials	5
Best Trial Number	1
Best Trial mAP@0.5	0.0377
Epochs per Trial	2
Final Training Epochs	3

Top 5 Trials (by mAP@0.5)

Trial	mAP@0.5	Hyperparameters
1	0.0377	{'optimizer': 'Adam', 'lr0': 0.0029743624956974353, 'lrf': 0.12274407820497062, 'm
0	0.0374	{'optimizer': 'Adam', 'lr0': 0.0010401663679887319, 'lrf': 0.1624074561769746, 'm
3	0.0367	{'optimizer': 'SGD', 'lr0': 0.0018235379940966055, 'lrf': 0.19739830637604217, 'm
4	0.0305	{'optimizer': 'Adam', 'lr0': 0.0020744284653995706, 'lrf': 0.11287713143501954, 'm
2	0.0143	{'optimizer': 'Adam', 'lr0': 0.0006675693250997204, 'lrf': 0.20016946012720685, 'm

Optimized Hyperparameters Used

Parameter	Value
optimizer	Adam
lr0	0.002974
lrf	0.122744
momentum	0.912982
weight_decay	0.000057
warmup_epochs	3
warmup_momentum	0.797665
warmup_bias_lr	0.096397
hsv_h	0.045387
hsv_s	0.783606
hsv_v	0.715865
scale	0.764245
translate	0.021880
mosaic	0.540900
mixup	0.064797
box	7.527204
cls	1.027950
df1	0.861551

Final Model Performance

Metric	Value
mAP@0.5	0.0339
mAP@0.5:0.95	0.0232
Precision	0.0415
Recall	0.0332

Generated by YOLO Training Notebook
BDD100K Dataset - Computer Vision Project